Commercial Value from Correlated Choice Models: Offer Optimization, Event Trading, and AI Service Routing

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Commercial research and validation plan

Abstract

Correlated choice models can support commercial decisions when dependence changes which action is profitable and the relevant outcomes can be measured. They do not, by themselves, establish a trading advantage, demand for a product, or an agent's absolute reliability. We develop three implementable proposals: contribution-profit optimization for merchants, relative-value trading in event contracts, and routing among AI services with verification and fallback. A sparse Gaussian identification result provides a possible way to learn substitution from controlled comparisons; its experimental assumptions and repeated-sampling cost are retained. We distinguish exact payoff relationships from speculative price convergence, and relative performance from successful task completion. For each proposal we specify the customer or counterparty, decision rule, required data, incumbent comparison, unit economics, and conditions for stopping. The provisional first business is a paid merchant experiment on an existing controlled sales channel. Prediction-market research begins with settlement semantics and executable costs; broad agent marketplaces and financial guarantees are deferred. Reproducible calculations quantify discount cannibalization, sample-size requirements, customer scale, trading capacity, and correlated fallback failures. These calculations are illustrative assumptions, not measured commercial results. No strategy backtest, customer pilot, or investment return is claimed.

Decision in brief. Obtain control of one economically valuable decision and its realized outcomes. Sell a measured improvement in that decision. Use the correlated model only if it improves net value against a simpler policy.

1 The commercial question

The relevant question is whether a decision-maker will pay for a measurable improvement in an existing activity. A merchant already chooses promotions. A trader already commits capital at quoted prices. An enterprise already pays for automated work and corrects failures. These activities have observable costs, incumbent policies, and people responsible for their budgets. They are suitable starting points for assessing the commercial value of choice models.

The research motivating this paper has two parts. Cotton (2021, 2026) study forward probabilities and inversion in stochastic contests: a joint performance model maps relative means to winning probabilities, and inversion fits relative means to specified probabilities under a chosen shock law. Wuebben (2026b) give an experimental identification result for a stable Gaussian factor model with idiosyncratic noise. Designed binary interventions identify the law of performance differences

under explicit conditions. Wuebben (2026a) explain applications and limitations through synthetic examples. The present paper asks what an operator must add before those results can support revenue or trading profit.

The commercial contribution is a set of decision specifications and falsifiable investment cases. It is not a new general theory of pricing or routing. In particular, the existence of a fast inverse does not establish that the true dependence structure is known, that a competitor is mispricing an event, or that a buyer will purchase a new service. A public mathematical result also cannot be treated as an exclusive commercial asset. A defensible business would need customer access, permission to collect useful outcomes, reliable execution, and an improvement large enough to cover acquisition and delivery costs.

1.1 A provisional ordering

Table 1 ranks the proposals by the feasibility of demonstrating economic value. This is a judgment conditional on access, not an estimate of market size. No merchant, enterprise design partner, trading account, or initial capital commitment is assumed to exist. The planning case is a founder-led team with one technical builder and part-time domain support. All financial amounts are in US dollars; they are not forecasts of local salaries or legal costs.

Proposal	First economically meaningful test	Provisional priority
Merchant offers	Randomized increase in realized contribution profit after returns, discounts and operating costs	First business if a merchant grants offer control and sufficient traffic
Event-contract trading	Positive net profit under exact settlement rules, conservative fills and capital charges	Bounded research project; deployment requires independent evidence of executable advantage
AI service routing	Lower total cost at an agreed quality and latency standard on a buyer's actual workflow	First business instead if enterprise data and a budget owner are accessible
Agent marketplace	Repeat transactions between existing buyers and providers with a reason to pay an intermediary	Expansion after a routing service has customers and repeat volume
Failure guarantees	Measured loss frequency, severity, dependence and funded contractual obligations	Defer until failure economics are established

Table 1: Access-dependent priorities. These are proposed tests, not completed experiments or market-size estimates.

The default recommendation is to sell a merchant profit experiment on a channel the merchant already controls. The first product need not depend on AI shopping traffic. If an enterprise workflow owner is substantially easier to reach, verified service routing can take its place. Running two full businesses before either has a paying customer would consume the same limited engineering and sales resources. A small prediction-market data study may be justified, but it has its own time and expense limit.

1.2 Evidence available at the drafting date

The two local research manuscripts contain synthetic calculations. They contain no measured merchant response, enterprise task-outcome panel, or live trading record. The present financial tables are generated from printed planning assumptions. Public documentation and primary research were checked on 6 September 2026 to establish that the relevant interfaces and competing approaches exist. Product announcements are evidence of infrastructure, not evidence of our ability to obtain access or earn a return.

That distinction changes the research sequence. A prototype demonstrating correct probability arithmetic can be useful, but the first investment decision should concern data and customer access. There is no basis here for an expected Sharpe ratio, a company valuation, a claimed addressable market, or a probability of commercial success.

2 What the choice model identifies, and what a business still needs

2.1 The experimental result

For a fixed task population or shopping context, consider latent utilities

$$U_i = \mu_i + v_i^T Z + \sqrt{d_i}\eta_i, \qquad Z \sim N(0, I_r), \quad \eta_i \sim N(0,1), \tag{1}$$

where $d_i \geq 0$ and $Z, \eta_1, \ldots, \eta_n$ are independent. The covariance is $\Sigma = VV^T + \mathrm{diag}(d)$. An alternative wins a menu when its utility is greatest. The model assumes that restricting the menu preserves the joint law of the retained utilities.

If a controlled intervention adds a known t to alternative i without changing the other utilities or shock law, its binary response against j is

$$q_{ij}(t) = \Phi\left(\frac{\mu_i - \mu_j + t}{s_{ij}}\right), \qquad s_{ij}^2 = \Sigma_{ii} + \Sigma_{jj} - 2\Sigma_{ij}. \tag{2}$$

Two interior population responses at $t_1 < t_2$ identify the mean difference and contrast variance. With $z_\ell = \Phi^{-1}(q_{ij}(t_\ell))$,

$$s_{ij} = \frac{t_2 - t_1}{z_2 - z_1}, \qquad \mu_i - \mu_j = s_{ij} z_1 - t_1. \tag{3}$$

For complete pair data, define $D_{ij} = s_{ij}^2$, $D_{ii} = 0$, and $P = I - \mathbf{1}\mathbf{1}^T/n$. Then $P\Sigma P = -PDP/2$ determines the Gaussian contrast law and consequently all menu and ranking probabilities on those alternatives.

The sparse result of Wuebben (2026b) assumes $r \geq 1$, $n \geq 3(r+1)$, positive pairwise contrast variances, and three disjoint groups of $r+1$ alternatives whose factor loadings are each affinely independent. Comparing every alternative with those reference alternatives identifies $P\mu$, PVV^TP and d, uniquely within factor dimension at most r. With $k = 3(r+1)$ references, the design has

$$m = kn - \frac{k(k+1)}{2} \tag{4}$$

distinct pairs. The reconstruction takes $O(nr^2 + r^3)$ arithmetic and $O(nr)$ storage in factor form. Local statistical stability additionally requires bounded parameters, positive idiosyncratic variances, informative treatment responses and uniformly conditioned reference groups. Unknown-rank recovery,

automatic selection of conditioned references, and unrestricted random-pair recovery with unknown diagonal variances are not established by that theorem.

The number of trials can dominate the arithmetic cost. With N independent observations at each of two levels per pair, there are $2mN$ binary observations. Table 2 illustrates this distinction for $r = 3$. The stated $N = 100$ is a counting assumption; it is not claimed to provide adequate accuracy for any commercial task. Reusing one sealed score draw at many offsets does not create independent new task outcomes. If full score vectors are available, their differences should generally be estimated directly.

Alternatives	Distinct pairs	Binary observations ($N = 100$)
1,000	11,922	2,384,400
1,000,000	11,999,922	2,399,984,400

Table 2: Observation counts under the minimum three-group design with twelve references. Two population probabilities per pair require repeated sampling.

2.2 Three gaps between identification and economic value

First, an observed discount h gives a utility shift βh only under an additive response with a common coefficient β. A common unknown positive coefficient permits a shared normalization, but unrestricted customer heterogeneity, attention effects and perceived-quality changes can invalidate a pooled Gaussian response. Randomization identifies the effect of the actual offer policy even when this structural interpretation fails. It does not make the structural assumptions true.

Second, financial trading does not ordinarily provide this intervention. Buying an election contract changes inventory and perhaps its price; it does not add a known amount to the underlying candidate's electoral performance. Calibrating a contest model from market prices therefore remains model-based inversion. The sparse experimental theorem does not identify electoral covariance from order flow.

Third, relative success and absolute success are different outcomes. This matters directly to the proposed agent exchange and its guarantees.

Proposition 1 (Rank information does not identify task completion)**.** *Suppose all observations are comparisons of $U_i + t$ with U_j, for arbitrary known offsets and menus. Without an absolute reference or additional outcome observations, these comparisons do not identify $\mathbb{P}(U_i \geq \tau)$ for a fixed externally defined success threshold τ.*

Proof. For any constant c, replacing U by $U + c\mathbf{1}$ preserves every comparison and every ranking, including all additive pair experiments. For a nondegenerate Gaussian coordinate, $\mathbb{P}(U_i + c \geq \tau)$ tends to zero or one as c tends to minus or plus infinity. The observations are unchanged along this family. □

This is an elementary consequence of the location indeterminacy, rather than a new identification theorem. A deterministic benchmark with an observed pass/fail outcome can supply additional information. Its validity must be established independently: adding a reference label to a leaderboard does not measure actual remediation cost. Common random shocks are likewise invisible to differences when added to every score and can change joint absolute failure probabilities. Correlation of latent ranking scores is not automatically correlation of operational failures.

2.3 When greater probability accuracy can pay

Let a denote an admissible offer or selection policy and Y its outcome. Suppose its realized contribution $g_a(Y)$ lies in an interval of width at most M, and let p_a and $\widehat{p}_a$ denote the true and fitted outcome laws. These distributions include the outside option when no purchase is possible. For discrete outcomes, our convention is $\mathrm{TV}(p, \widehat{p}) = \frac{1}{2}\sum_y |p(y) - \widehat{p}(y)|$.

Proposition 2 (A sufficient decision-accuracy bound)**.** *If* $\mathrm{TV}(p_a, \widehat{p}_a) \leq \epsilon$ *for every action in a fixed finite action set, choosing an action with the largest fitted expected contribution loses at most* $2M\epsilon$ *in true expected contribution relative to the best action in that set.*

Proof. For a function with range width at most M, the difference between its expectations under two distributions is at most M times their total variation distance. Let a^* maximize true value and $\widehat{a}$ maximize fitted value. Insert the fitted values of these two actions between their true values. The fitted-value difference is nonpositive in the regret calculation; the two remaining errors are each at most $M\epsilon$. □

The bound is a decision-theoretic diagnostic, not a commercial guarantee. Actual profit can have refund and loss tails beyond the assumed interval. Moreover, the local identification paper's fixed-menu bound does not provide uniform error control for arbitrary prices, contexts or changing menus. With $M = 100$ and $\epsilon = 0.01$, the bound permits a loss of \$2 per decision, which can exceed the improvement being pursued. Reporting lower probability error is consequently insufficient. The operational test is incremental profit or total cost on future decisions.

3 The first business: merchant contribution-profit optimization

3.1 Buyer, problem and initial product

The initial buyer is a head of e-commerce or pricing, with a finance owner who can approve the profit measurement. The qualifying merchant sells several substitutable products, can vary offers on an owned channel, has reliable contribution-cost data, and has enough eligible traffic to run a powered test. The channel can be a product comparison page, a quote flow, or a merchant-owned shopping assistant. A store selling only a few high-value items each week is unlikely to generate the needed experimental evidence quickly.

The proposed product returns one approved offer from a small discrete set, subject to merchant-specified margin and inventory constraints. It records the decision, follows the resulting order through returns, and produces a finance- reconciled treatment-control comparison. Its sales proposition is a measured increase in retained contribution profit. Conversion alone is a misleading target when incentives are costly or sales move away from higher-margin items.

The first implementation should use a merchant's existing experimentation and checkout systems wherever possible. This is already a competitive market. Intelligems advertises price, discount, shipping and checkout experiments with cost-of-goods-based metrics (Intelligems, 2026). The commercial comparison is therefore against the merchant's current policy and a competent experiment-led alternative, not against a fictional operator who discounts at random forever. The correlated model earns a place only if cross-product substitution improves decisions beyond those alternatives.

3.2 The profit objective

Let x contain information available before the offer: eligible product set, stock, customer or session attributes that may be used, delivery constraints, and time. Let $S(x)$ be the observed choice set, with 0 denoting no purchase within the prespecified attribution window. For products owned by the merchant, let $M \subseteq S(x)$ and let d be an admissible discount vector. Define

$$V(d, x) = \sum_{i \in M} p_i(d \mid x, S)\overline{m}_i(d, x) - c_{\text{decision}}(d, x). \tag{5}$$

Here $\overline{m}_i(d, x)$ is expected retained contribution conditional on buying i under that policy. It includes collected revenue less discounts, product cost, payment and channel charges, shipping subsidies, expected returns, chargebacks and other incremental servicing costs. The decision cost includes charges incurred even without a purchase. Estimation should use the realized transaction ledger as the outcome rather than assume these conditional margins are policy-invariant.

The policy solves $\max_{d \in \mathcal{D}(x)} V(d, x)$, with an unchanged offer included in $\mathcal{D}(x)$. Floors on retained margin, offer ceilings, stock limits and a restricted action set are explicit business constraints. For a single product with constant pre-discount contribution m and differentiable purchase probability $p(d)$, an interior stationary point satisfies

$$(m - d)p'(d) = p(d). \tag{6}$$

This is only a first-order condition. Without additional shape restrictions, it proves neither uniqueness nor global optimality; the finite approved action set can instead be evaluated directly.

For multiple products with fixed pre-discount margins and a common utility response β to money, let $L_{ij} = \partial p_i / \partial \mu_j$. Then, locally and with other discounts held fixed,

$$\frac{\partial V}{\partial d_j} = -p_j + \beta \sum_{i \in M} (m_i - d_i)L_{ij} - \frac{\partial c_{\text{decision}}}{\partial d_j}. \tag{7}$$

The sum makes substitution economically relevant. A discount can divert buyers from a high-margin product owned by the same merchant. For smooth additive random utility, competing alternatives have nonpositive cross derivatives; the magnitude and ownership of diverted demand matter. If margins, return rates or service costs depend on the offer, their derivative terms must also be included. Equation (7) is a restricted explanatory calculation, not the proposed estimator for an unrestricted transaction ledger.

Discount A	p_A	p_B	No purchase	Conversion	Contribution
\$0	0.040	0.060	0.900	10.0%	\$8.400
\$5	0.065	0.052	0.883	11.7%	\$8.775
\$10	0.090	0.030	0.880	12.0%	\$7.500

Table 3: Illustrative discount cannibalization. A contributes \$60 before its discount and B contributes \$100. Probabilities are supplied scenario inputs, not data or a fitted Gaussian demand system. Contribution is per eligible session, before any common decision charge.

Table 3 gives a simple reason to optimize the merchant's full contribution. A \$10 discount raises conversion from 10% to 12%, yet lowers expected contribution from \$8.400 to \$7.500 per session. A \$5 discount gives \$8.775 and is best among the three supplied actions. The example demonstrates an accounting possibility, not a measured discount response. It establishes neither Gaussianity nor a likely magnitude of attainable uplift.

3.3 What can actually be observed

A merchant's order feed records its own sales, not the losing offers at every competitor. Nor does a completed checkout reveal every product that a buyer considered. Accordingly, the first experiment uses a controlled, logged merchant menu. If competitor offers are missing, a competitor-specific substitution matrix is not claimed. No purchase is an observed outcome within the merchant channel; it does not distinguish postponement, abandonment, or purchase elsewhere without additional data.

The simple winner model concerns one choice. A multi-item basket, subscription renewal, stockpiling response, or discount that changes future shopping requires a different outcome model. The first pilot should restrict the setting or evaluate the complete basket and repeat-purchase contribution directly. It should not manufacture a single-choice likelihood from a basket transaction.

Prices in a scraped catalog likewise do not establish what the customer saw. The required menu log records actual exposures, prices, inventory and delivery terms at the decision time. When that log cannot be obtained, a randomized offer-policy test may still identify the merchant's overall effect, but the structural account of who substituted for whom remains limited.

3.4 AI commerce is a distribution option

Google's Universal Commerce Protocol describes checkout capabilities and extensions, including discounts (Handa and Gupta, 2026). Google's May 2026 announcement describes an expansion of its Direct Offers pilot (Google, 2026). These interfaces make merchant offer logic a relevant application, but neither announcement establishes open access to a real-time auction over a buyer's full consideration set.

OpenAI's September 2025 announcement introduced the Agentic Commerce Protocol with Stripe (OpenAI, 2025). It should be read as a dated launch announcement. Shopify's June 2026 description distinguishes channel-specific checkout paths, including merchant-site checkout through an in-app browser for ChatGPT (Shopify Staff, 2026). The integration route must therefore be confirmed for the chosen merchant and channel. A protocol specification does not guarantee that every checkout customization or randomized offer can be applied there.

The practical first release works on the merchant's existing site or quote flow. An AI channel is added when its documented interface supports the needed offer action, exposure log and outcome attribution. This avoids making early revenue depend on an unconfirmed platform permission. Public growth claims about AI referrals are not converted into a serviceable market estimate here.

4 Measuring merchant value and earning the fee

4.1 A prospective experiment

The pilot has four stages: instrumentation, randomized exploration, policy selection using only accumulated training data, and a separate prospective comparison of the selected policy with the incumbent. An A/A period first checks that assignment, exposure, orders and returns reconcile without an artificial treatment difference. The same customer receives a consistent assignment where cross-session contamination is plausible. Inventory interference may require randomization across suitable time blocks or product groups instead; its design and uncertainty calculation must match the unit of randomization.

The primary outcome is retained contribution per eligible randomized unit, including units

with zero purchases. The denominator is set before treatment. Order cancellations and returns are included after an agreed maturity window. Changing the denominator to purchasers would condition on a treatment outcome. Total contribution across the owned substitute set is measured, so a gain on one product cannot conceal losses on another.

The policy comparison includes: the existing offer rule, a simple segmented best-offer policy, a logit or other standard response model where appropriate, and the correlated model. Models receive the same information available before the decision. If the correlated model does not increase prospective net profit after its additional computational and maintenance costs, it is removed from the commercial product. A profitable simple policy remains a legitimate business result.

Exploration records the probability $e(a \mid x)$ of every assigned action. For a fixed candidate policy π, one possible supplementary offline estimator is the doubly robust policy-value estimator (Dudík et al., 2011):

$$\widehat{V}(\pi) = \frac{1}{T}\sum_{t=1}^{T}\left[\sum_{a}\pi(a \mid x_t)\widehat{m}(x_t, a) + \frac{\pi(a_t \mid x_t)}{e(a_t \mid x_t)}\{Y_t - \widehat{m}(x_t, a_t)\}\right]. \tag{8}$$

Here Y_t is realized contribution and $\widehat{m}$ is a fitted conditional mean. This requires consistency of outcomes, action overlap and a defensible assignment mechanism; nuisance predictions and policy selection are separated from their evaluation observations. An unoffered action cannot be evaluated by assigning it an invented propensity. Cluster dependence and adaptive assignment require corresponding inference. The final customer claim is based on the prospective policy comparison, not the best-looking value after searching many policies on the same log.

4.2 Traffic can be the binding constraint

For a prespecified comparison of two equally sized independent groups with common outcome standard deviation σ, a normal approximation to 80% power at a two-sided 5% level gives

$$n_{\text{arm}} \simeq \frac{2(z_{0.975} + z_{0.80})^2\sigma^2}{\Delta^2}. \tag{9}$$

This is a planning calculation, not an exact test for heavy-tailed profit data. Table 4 sets $\sigma = \$26$, approximately the standard deviation of the unchanged-offer scenario in Table 3. A pilot must replace this assumption with an estimate from its own historical or A/A data.

Effect/session	Observations/arm	Total observations
\$0.100	1,061,169	2,122,338
\$0.250	169,787	339,574
\$0.375	75,461	150,922

Table 4: Illustrative sample sizes for a fixed two-policy comparison, 80% power, two-sided 5% level and \$26 standard deviation per independent unit. These do not include policy-learning samples or a multiple-comparison penalty.

Detecting \$0.10 per session requires over two million independent observations in this scenario. Detecting the illustrative \$0.375 improvement requires 150,922. If only 10% of store traffic is eligible, total site traffic is not the available sample size. Repeated users, seasonal changes, return delays and inventory dependence further affect feasibility. Small pilots should target a larger measurable

business decision or run longer; they should not describe an underpowered null result as evidence of no effect.

The selling threshold is economic as well as statistical. Before the comparison, the parties specify the smallest benefit that covers the customer's switching and operating costs and the supplier's fee. A lower confidence bound above that threshold is the proposed rollout condition. A nonsignificant estimate triggers the previously agreed maximum-duration or maximum-budget decision, rather than an indefinite sequence of new segment searches.

4.3 Commercial terms and attribution

The first contract can charge a fixed \$10,000–\$25,000 instrumentation and analysis fee, explicitly offered as a negotiating hypothesis. This is a paid service deliverable and does not promise a positive treatment effect. Payment willingness is untested. A portion can be credited against a later service contract if the pilot produces a finance-approved benefit.

For a performance component, let $\Delta\Pi_H$ denote estimated incremental contribution over an agreed measurement period H. A possible fee is

$$F_H = \tau \max\{0, \underline{\Delta\Pi}_H\}, \tag{10}$$

where $\underline{\Delta\Pi}_H$ is a prespecified conservative estimate, such as a valid lower confidence bound, and τ is the negotiated share. The measurement schedule and handling of repeated looks are fixed in advance. Loss carryforwards and periodic reconciliation prevent the supplier from collecting on favorable noisy periods while discarding unfavorable periods. An alternative is a fixed recurring fee following a successful experiment; value-based pricing is not mandatory.

The agreement must define eligibility, assignment, returns, horizon, baseline changes, taxes, incremental servicing costs and audit rights. For a subscription product, retaining a randomized control fraction can support continued measurement, subject to commercial and power considerations. Annual revenue projections must use the traffic actually receiving the policy, not all store sales. Training and measurement costs, as well as the foregone gains of the control group, belong in the total business case.

4.4 Customer economics and supplier scale

Let G be annual baseline contribution on eligible traffic to which the new policy is applied, u its fractional improvement, and τ the effective fee share actually earned after measurement rules. A simplified mature account has

$$R = \tau u G, \qquad \Pi_{\text{supplier}} = J(R - c) - F, \tag{11}$$

where J is the customer count, c annual direct delivery cost per customer, and F annual fixed operating cost. This expression excludes one-time pilot fees and assumes collected fees, stable accounts and no bad debt. In an actual plan, onboarding, churn, acquisition expense and collection timing are modeled explicitly. A statistically conservative fee can make realized R lower than the deterministic scenario.

At the 3% assumption, the customer gains \$300,000 before the fee and retains \$255,000 afterwards, before its additional internal costs. The supplier earns \$45,000 revenue and \$33,000 direct contribution per customer. Twenty such customers produce \$900,000 revenue but only \$60,000 operating profit after the assumed fixed expense. Forty produce \$720,000 operating profit.

Uplift	Annual fee	Direct contribution	Break-even clients	Clients for $1m
1%	$15,000	$3,000	200	534
3%	$45,000	$33,000	19	49
5%	$75,000	$63,000	10	26

Table 5: Illustrative mature-account economics: $G = \$10$ million, $\tau = 15\%$, $c = \$12,000$ and $F = \$600,000$ annually. The last column targets $1 million annual operating profit before tax, with customer counts rounded up. The uplifts are assumptions.

Forty-nine are needed to exceed $1 million. This is a possible valuable company, but it requires many substantial eligible accounts or materially better economics.

At $45,000 per account, $10 million recurring revenue requires 223 equivalent accounts and $100 million requires 2,223. These are arithmetic requirements, not a count of available buyers. Qualifying merchants may be much scarcer than the total merchant population because the eligible traffic and contribution base are restricted. At an assumed 25% contribution margin, the $10 million eligible contribution base requires $40 million eligible annual sales. If only 10% of sales are eligible, total sales would need to be $400 million under the same margin assumption. The commercial segment must therefore be chosen from eligible economics rather than a generic list of small online stores. At a 1% improvement, direct contribution falls to $3,000 per customer, making the same delivery model unattractive.

Customer acquisition can overwhelm the model as well. A hypothetical $25,000 acquisition and onboarding cash outlay takes about 9.1 months to recover from $33,000 annual direct contribution, assuming revenue starts immediately and the account remains active. A three-month sales and deployment delay extends cash recovery accordingly. An assumption that direct support costs only $12,000 must therefore be tested early. The founder should log implementation hours and support demands from the first pilot.

5 Prediction-market trading: a separate investment case

5.1 The source of possible profit

Three different activities must be distinguished. Exact payoff arbitrage buys a package whose contractual payoff is bounded below by more than its full cost. Statistical relative value trades a forecast of future prices or outcomes and can lose. Market making supplies immediacy, earning spreads and possibly rebates while taking inventory, adverse-selection and operating risk. They require different evidence and should have separate profit attribution.

There is existing research on prediction-market arbitrage. Saguillo et al. (2025) analyze historical Polymarket inconsistencies and extracted profits. That establishes prior analysis and competition; it does not establish current accessible opportunities or the performance of the strategy proposed here. The appropriate first comparison is a semantics-correct payoff scanner with conservative execution costs. A Gaussian contest model is unnecessary for identifying an exact relationship between contractual payoffs.

5.2 Begin with a small, auditable outcome space

Choose one venue and one family of contracts referring to the same observable quantity, measurement source and resolution time. A family of threshold and interval contracts on a single reported scalar is a suitable research target if the definitions can be verified. The initial family is selected before examining its apparent profit. Different sources, revised versus first releases, time zones, missing observations, cancellation and exceptional resolution clauses may destroy an apparent equivalence.

Partition all contractually possible outcomes into a finite set Ω. Let $A_{\omega j}$ be the terminal dollar payoff of one purchasable instrument j in state ω. YES and NO instruments have their own columns. For a nonnegative purchase vector x and full acquisition-and-carry cost $K_t(x)$, the guaranteed terminal surplus of a completely acquired package is

$$g_t(x) = \min_{\omega \in \Omega} (Ax)_\omega - K_t(x), \qquad x \geq 0. \tag{12}$$

Displayed depth, lot sizes, cash and position limits constrain x. At fixed price levels and a suitable linear fee representation, maximizing this surplus can be formulated as a linear program; discrete lots and nonlinear costs can require a different optimization. Actual venue costs determine the calculation. An optimizer does not establish that the state list or settlement definitions are complete.

For a mutually exclusive exhaustive partition, buying one YES share of each outcome pays one dollar in every state. For two events satisfying $A \subseteq B$, buying YES on B and NO on A gives

$$\mathbf{1}_B + (1 - \mathbf{1}_A) \geq 1. \tag{13}$$

The three possible $(\mathbf{1}_A, \mathbf{1}_B)$ states $(0,0), (0,1), (1,1)$ pay $1, 2, 1$, respectively. This uses purchasable long instruments and does not assume unrestricted short selling. The impossible state $(1,0)$ would pay zero: if contract wording permits that state, the proposed lower bound fails.

The cash flows must be comparable in maturity and collateral terms. A void refund may have a different payoff from a binary win. Cross-venue packages add separate oracle, counterparty, withdrawal and settlement risks and may require collateral at both venues without netting. They are a later research stage. An automated language model can suggest related contracts, but the trading relationship is admitted only after a human-readable payoff map and versioned rule references have been checked.

5.3 Fees can consume the entire discrepancy

Kalshi documents market-data and execution APIs, including a public-data quick start (Kalshi, 2026a,c). Its fee document effective 7 July 2026 gives a general taker formula proportional to $0.07MCP(1-P)$, with contract multiplier M, share count C and price P, and lists exceptions and maker charges (Kalshi, 2026d). The document's rounding text and displayed examples should be reconciled against the venue's current calculator and actual fill ledger before production use. A historical backtest needs the schedule applicable at the historical trade time.

As an arithmetic illustration, YES and NO asks of \$0.49 each create a gross \$0.02 package surplus. Applying coefficient 0.07 to both legs before rounding gives fees of

$$2(0.07)(0.49)(0.51) = \$0.034986. \tag{14}$$

The package then loses about 1.50 cents before additional costs. This is not an observed quote or a venue invoice. It demonstrates why a two-cent discrepancy cannot be judged without its fee and execution context.

Polymarket's documentation likewise distinguishes fee-bearing categories and fee parameters (Polymarket, 2026). Neither venue is represented here by a permanent universal fee assumption. Rebates and temporary rewards are excluded from the base case and reported separately when actually earned. The model also includes bid–ask costs, depth, settlement funding, failed hedges and fees of any intermediary used to access the venue.

5.4 A completed package and an attempted package are different

Equation (12) applies to the acquired package. Independent fills of several limit orders are not an atomic transaction. Let g be the net gain if every leg fills, ℓ the loss if an incomplete package must be unwound, and f the probability of that failure. A simplified attempt has expectation

$$(1 - f)g - f\ell. \tag{15}$$

With supplied inputs $g = \$0.006$, $f = 5\%$ and $\ell = \$0.15$, this is $-\$0.0018$ per attempt. The numbers are illustrative, but the loss mechanism is central. Order size is constrained by the weakest leg, measured cancellation latency, and the ability to terminate or unwind exposure.

A taker simulation consumes the contemporaneously available depth after the assumed processing and network delay. A maker simulation records queue position or an explicitly conservative proxy and requires evidence of volume trading through that position. Touching a displayed price is not sufficient evidence of a fill. If the data cannot identify queue outcomes, report execution bounds or collect better data; do not assign favorable fills by default.

Kalshi's documentation separates live and historical endpoints for markets, trades and candlesticks (Kalshi, 2026e). Their existence is not proof of a complete historical depth-and-queue record. Current order books document available bid levels and the YES/NO complement convention (Kalshi, 2026b). A production research log needs feed sequence integrity, receive timestamps, clock checks and observed outages. Permission to use and retain the selected feed is a separate deployment prerequisite.

5.5 A statistical relative-value strategy

Only after the payoff and execution baseline is functioning should a stochastic model be evaluated. Let Y be a vector of terminal contract payoffs and $\widehat{q}_t = \mathbb{E}_{\widehat{P}_t}[Y \mid \mathcal{F}_t]$ a forecast formed from information available at time t. Here $\mathcal{F}_t$ contains only data that had arrived by the decision time. A family of mutually exclusive contestants can use a Gaussian race. Overlapping binary events require a joint state model; their probabilities cannot be forced to sum to one as if only one could occur.

For a joint finite-state model with probabilities π_t, marginal values are $q_t = A^T \pi_t$. Recovering π_t from marginal prices may be underidentified. A regularizer or a Gaussian structure can select one completion but cannot turn that selection into observed joint information. In the existing companion, two covariances fit the same first-place probabilities but imply different ranked-contract values. That is model uncertainty to price and test.

For a long single contract held to resolution, a conservative research rule is to consider purchase only when

$$\inf_{P \in \mathcal{P}_t} \mathbb{E}_P[Y_j \mid \mathcal{F}_t] - \mathrm{ask}_t(j, q) - c_t(j, q) > b_t(j, q). \tag{16}$$

The ask is the volume-weighted executable acquisition price for quantity q, c_t includes fees and funding not already in the ask, and b_t is a prespecified reserve for estimation and execution error.

The set $\mathcal{P}_t$ contains models or parameter regions estimated from prior data. Unless it has validated statistical coverage, the infimum is a stress test, not a confidence bound. A large model uncertainty set may correctly produce no trade.

For convergence trading, the target changes to the future executable exit price at a specified horizon. A current residual to a fitted surface supplies no evidence that the residual will mean-revert. The study must prespecify its exit horizon, maximum holding time, invalidation rule and news exclusions, then estimate future net liquidation value on chronological holdouts. Holding to settlement and exiting on convergence have different capital and loss profiles; their results are reported separately.

The key baselines are current executable prices, a simple calibrated marginal forecast, and a low-dimensional event model without the proposed correlated contest structure. A comparison that feeds the target's own current price into an inversion and then calls the reconstructed target an independent forecast is circular. Leave-target-out calibration can prevent this particular defect, but it still needs genuine future-outcome or future-price validation.

5.6 Market making needs an order-flow model

For a specified Gaussian contest, the Jacobian $L = \partial p / \partial \mu$ describes how model probabilities respond to changes in latent means. The identity $dp = L\, d\mu$ does not determine what latent update follows a trade. A trade might carry information, hedge an existing position, or express an uninformative liquidity demand. The update

$$d\mu = H_t(\text{order flow, news, depth, prior state}) \tag{17}$$

is an additional statistical model to be fitted and tested.

Inventory risk also concerns payoffs rather than the covariance of Gaussian latent scores. For one exhaustive winner vector with probabilities p, the payoff covariance is $\operatorname{diag}(p) - pp^T$. For overlapping events it is instead $\mathbb{E}[YY^T] - \mathbb{E}[Y]\mathbb{E}[Y]^T$ under a coherent joint law. These are different from Σ in (1). Confusing them would produce a faulty hedge.

For illustration, an inventory-adjusted reservation value may subtract the marginal increase in a chosen convex risk charge $\mathcal{R}$:

$$r_j(t, I) = \widehat{q}_{j,t} - \{\mathcal{R}_t(I + e_j) - \mathcal{R}_t(I)\}. \tag{18}$$

Bid and ask distances from that value must cover conditional adverse selection, fees, uncertainty and the empirically estimated fill process. A positive spread is not automatically profitable. Markouts after fills, open inventory losses, rebates and terminal settlements need separate ledgers. A new operator should not begin with autonomous quoting across thousands of contracts before those quantities can be reconciled.

5.7 Research protocol, capacity and stop conditions

The first trading research release covers one rule-reviewed family and one venue. It records metadata, available depth and trades, with a dated fee table. It evaluates exact packages first and model-based opportunities separately. The evaluation split is chronological and groups all linked contracts from an event together. Training windows, normalization, factor selection, contract relationships and model tuning use only information available before the test decision. Revised data and late-arriving news

use their actual availability times. Repeated ticks from one event are not independent investment trials.

The performance report includes realized and unrealized net profit, costs by type, maximum drawdown, average and peak capital tied up, exposure to each underlying event, failed-package losses, return on committed capital, and profits by event and date. It reports the number of independent event clusters and all attempted model variants. A Sharpe ratio, if eventually reported, needs an appropriate time-series frequency and treatment of autocorrelation; it is not constructed from the apparent sample size of quotes.

The initial protocol reserves a completely untouched final event/date block. A result driven by one resolved event, disappearing under observed latency, or requiring unrecorded fills does not justify deployment. The proposed deployment condition is positive net value under a prespecified conservative cost case, positive evidence across independent event clusters, and sufficient capacity to cover operating expense. Thresholds and the data horizon are fixed before examining final results; insufficient events produce an inconclusive study, not a passed strategy.

Let Q count completed normalized strategy units per year, with entry, hedge and exit legs counted once as a complete economic position. Let e be expected net dollars per unit after all variable costs, and F_T fixed annual expense. Then the planning requirement for target profit Π_T is

$$Q = \frac{\Pi_T + F_T}{e}. \tag{19}$$

The definition prevents multiplying a per-package edge by exchange volume that counts both sides or each leg several times.

Net edge	Completed units/year	Addressable units/year	Cash capacity
\$0.002	650m	65bn	\$3,250,000
\$0.005	260m	26bn	\$1,300,000
\$0.010	130m	13bn	\$650,000

Table 6: Illustrative scale needed for \$1 million annual operating profit with \$300,000 fixed expense. Addressable activity assumes 1% participation in comparable units; cash capacity assumes 100 annual turns and \$0.50 tied per unit. None of these capacity inputs has been measured.

At half a cent net edge, the requirement is 260 million completed units per year. The illustrative cash requirement is \$1.3 million at 100 turns, but \$13 million at ten turns with the same cash per unit. Many multi-leg packages need more than \$0.50 per normalized unit, and peak inventory can exceed its average substantially. Capital may be partitioned across venues. A large headline market volume does not establish that this strategy can trade the required volume at the assumed edge.

Before live deployment, the actual legal entity, residence, venue eligibility, contract permissions, data rights and position limits must be checked under the then-current rules. This paper does not assume account access. The initial task is a data and execution study; funding an account or placing orders is a separate investment decision after the study.

6 AI services: verified routing before a marketplace

6.1 Choose a narrow workflow with observable outcomes

The candidate buyer is an operations owner with a repeated task and a measurable completion standard. One possible first workflow is invoice extraction and reconciliation that produces a proposed record for human or deterministic verification, rather than autonomously approving payments. The task definition specifies fields, source documents, matching rules, exceptions and the audit record. A financial-document extraction or software maintenance workflow with equally observable outcomes could serve instead.

The first product compares a small number of already procured services, routes eligible tasks, verifies outputs, and escalates failures. It does not need an open exchange, a token, a universal agent score, or thousands of vendors. The buyer pays for lower verified total cost under quality and latency constraints. The budget owner must supply representative task data and accept the definition of success before model development.

Generic routing already has established alternatives. RouteLLM learns model routing from preference data (Ong et al., 2024); Amazon Bedrock documents a commercial intelligent prompt-routing service (Amazon Web Services, 2026). The proposed distinction would have to be task-specific outcome verification and economically useful selection of alternative providers, including failure dependence. It is not the mere existence of a router. These sources are evidence of related methods and products, not an exhaustive competitor survey or a claim that no incumbent can supply the proposed workflow.

6.2 The objective and the role of dependence

For task features x known before dispatch, define total cost

$$L_i(x) = \text{service cost} + \text{verification cost} + \text{delay cost} + \text{remediation and downstream loss}. \tag{20}$$

A one-provider rule minimizes $\mathbb{E}[L_i \mid x]$ over eligible services subject to the buyer's quality and latency requirements. Marginal conditional costs are enough for that decision. Cross-provider covariance need not improve it. Furthermore, the provider most likely to rank first can have a worse expected loss because winner frequency ignores the magnitude of losses.

Dependence becomes useful when the action includes multiple providers, inspection or a fallback. Consider a restricted sequence: run A, detect every failure perfectly, run B if A fails, then pay a fixed remediation cost R if both fail. Let F_A, F_B be the failure events on the same task, c_A, c_B the attempt costs, and $p_A = \mathbb{P}(F_A \mid x)$. Under this specific information structure,

$$\mathbb{E}[L_{A\to B} \mid x] = c_A + p_A c_B + \mathbb{P}(F_A \cap F_B \mid x)R. \tag{21}$$

The B outcome must refer to the actual fallback condition, including any context passed from A. Reusing B's unconditional pass rate among ordinary tasks would generally be incorrect.

The first row of Table 7 corresponds to independent failures. The second has strongly overlapping failures. Expected total cost rises from \$0.56 to \$2.96 despite unchanged marginal failure rates. This is the commercial quantity a complementary-provider model could improve. It is measured from paired task outcomes and the actual fallback mechanism, not inferred from marginal leaderboard rankings.

Perfect verification is an intentionally strong simplification. An operational model needs false acceptances, false rejections, timeouts, correlated outages and delay to escalation. For example, an

Joint failure	Failure of B given A failed	Expected cost/task
1%	10%	\$0.56
9%	90%	\$2.96

Table 7: Illustrative fallback economics with marginal failure probabilities 10% for both providers, $c_A = \$0.20$, $c_B = \$0.60$ and $R = \$30$. The two joint failure laws satisfy the same marginal rates.

undetected error can bypass fallback and incur downstream loss. A router that saves model fees by increasing such errors can be much more expensive. Tasks should initially remain within a workflow where errors are observable before expensive irreversible actions.

6.3 Evaluation data and commercial terms

The minimum dataset links task features, timestamped provider and model version, prompt and tool configuration, charged cost, latency, outputs, verification results, eventual corrections and reviewer effort. The outcome standard is agreed before evaluation. A subset of tasks is evaluated by multiple providers to estimate joint failures. Paired tasks, organization-level shifts and provider versions are held together in the evaluation split. Updating a model or verification rule creates a new relevant regime.

Task-conditioned Gaussian scores are a possible comparative model. Their feature map, score scale, rank and stability must be justified separately; a linear expression $\mu_i(x) = x^T\theta_i$ does not prove those properties. If cardinal scores and completion labels are observed, use that information directly. Sparse winner-only experiments are relevant only when the interface withholds scores and allows valid offsets. Artificially reducing rich logs to binary wins would discard information the buyer already paid to obtain.

The prospective comparison is against the cheapest acceptable single provider, the buyer's incumbent, and a simple validated escalation rule. All costs in (20) enter the primary outcome. Predeclared noninferiority margins for serious errors and latency prevent a low-cost policy from passing by silently degrading quality. The trial needs sufficient independent tasks in the risk-relevant strata, not just an aggregate benchmark score.

A paid implementation and evaluation project is a possible first sale. A recurring fee can then be capped as a share of verified savings. For example, one million annual tasks, \$0.30 verified net savings per task and a 20% share yield \$60,000 annual supplier revenue. These are assumptions, not a quoted market price. The economics are plausible only if bespoke integrations, human auditing, security review and ongoing support cost substantially less. At much lower task volume or savings, a separately sold router may be a feature of another product rather than a viable stand-alone company.

6.4 Why underwriting and an exchange come later

An agent's probability of ranking first cannot price a failure guarantee, as Proposition 1 shows. Even an observed pass rate needs a loss model. Expected covered loss is frequency times conditional severity, with additional expense and capital required for correlated incidents. A minimal premium calculation is

$$\text{premium} \;\geq\; \mathbb{E}[\text{covered loss}] + \text{administration} + \text{capital charge} + \text{uncertainty allowance}. \tag{22}$$

Calling the promise a service credit does not establish that every possible contract lies outside applicable insurance or other rules. Its legal treatment depends on actual obligations and jurisdiction and must be resolved before such a product is offered. An initial bounded remediation service can instead be costed as an operational activity with explicit limits.

Rare-event evidence is expensive. For N independent, identically distributed trials with zero failures, an exact one-sided 95% binomial upper bound is

$$q_U = 1 - 0.05^{1/N}. \tag{23}$$

At least 373 zero-failure trials are needed to put that bound at or below 0.8%; at least 2,995 are needed for 0.1%. These are best-case counts at a fixed sample size, before task heterogeneity, provider changes, multiple claims or common incidents. They provide no estimate of loss severity and no evidence about rare catastrophic failures absent from the task sample.

An open marketplace adds provider acquisition, buyer acquisition, identity, dispute resolution, payments and an incentive to transact through the platform again. Reputation research recognizes task heterogeneity and evaluation incentives (Chishti et al., 2026); that is not evidence that a new exchange has demand. The service should expand into marketplace functions only after existing customers repeatedly request provider discovery or procurement and will pay for it. A take rate must compensate a continuing service, otherwise repeat buyers and providers have a reason to contract directly.

7 Execution as a business plan

7.1 Access determines the first product

The immediate decision is which party will provide an executable pilot. A qualifying merchant supplies offer control, exposure logs, cost accounting and sufficient traffic. A qualifying enterprise supplies representative tasks, verification outcomes, provider access and an accountable budget owner. Institutional trading relationships may instead make a dealer-inquiry product more accessible. Cotton's original inquiry application selects dealers while accounting for inquiry costs (Cotton, 2021); the existing companion explains the dependence benefit.

A dealer-inquiry service would require timestamped invitations, responses, prices, no-quotes, sizes, inventory context and executions, including outcomes for losing dealers. Public transaction prints alone do not reveal the counterfactual offers of uninvited dealers. The decision objective would be execution value after inquiry and information-leakage costs, not an inferred dealer ranking. This is a conditional alternative if a desk can supply the required logs and support a controlled evaluation. No such relationship is assumed in the present plan.

The potential institutional economics can be expressed without inventing an execution result. If a desk routes annual notional V and a controlled study finds improvement b basis points in its all-in execution cost, measured in dollars per dollar of notional, annual savings are $Vb/10{,}000$. With assumed $V = \$50$ billion and $b = 0.2$, savings are \$1 million; a 15% fee would be \$150,000 before supplier costs. This illustrates why a few large workflows can support a service. The improvement, routed volume and willingness to pay remain unmeasured, and the comparison must include no-quotes, lost fills, latency and the cost of revealing trading interest. A change in dealer selection without a credible counterfactual execution benchmark is insufficient.

The first two weeks should therefore establish whether a party will supply the essential data and pay for a defined evaluation. Expressions of interest without access or a budget owner are

weak evidence. Building a universal platform before resolving this question is not justified by the mathematics.

7.2 A six-month capital allocation

Table 8 supplies one explicit budget for the main business experiment. It includes a founder draw and external technical capacity; it is a planning envelope rather than a market salary survey. The founder may be able to reduce cash costs, but unpaid labor still has an opportunity cost. The budget excludes trading capital, insurance reserves and customer-funded discounts. Merchant incentives and support costs are nevertheless included in the customer's measured economic outcome.

Six-month expense assumption	Amount
Founder draw	\$60,000
Engineering and data	\$80,000
Experimentation and compute	\$15,000
Customer integration and domain support	\$20,000
Contracts and security	\$15,000
Contingency	\$10,000
Total	\$200,000

Table 8: Illustrative six-month cash budget for one main business pilot and initial product. No customer revenue is assumed to finance it.

The \$200,000 envelope is released by milestones. At most \$20,000 is used to establish access, data feasibility and a written pilot scope. The next \$40,000 funds instrumentation, baseline measurement and a functioning experiment. A further \$60,000 funds the prospective evaluation after those steps succeed. The remaining \$80,000 funds productization and a second customer only if measured value and delivery economics justify it. These tranches sum to the budget; the release conditions matter more than the calendar. Failure to satisfy them preserves the unspent balance.

An optional prediction-market study is capped separately at \$20,000 and four to six weeks of research effort. It must not consume the main customer's implementation capacity. The amount is research expense, with no trading balance included. A capital allocation for actual trading is specified only after fill, capacity and loss evidence exist. Launching both a trading desk and a software company within the same \$200,000 assumption is not proposed.

7.3 Milestones and decisions

Period	Required deliverable	Investment decision
Weeks 1–2	A proposed list of 15 relevant buyer conversations; at least one budget owner supplies a representative data extract and agrees a priced pilot scope, measurement definition and access path	Without qualified access, change the customer segment or pause before building the optimizer. The conversation target is a plan, not a completed sales funnel.
Weeks 3–6	Cost ledger, exposure or task log, A/A or equivalent pipeline check, simple baseline, sample-size estimate, and maximum trial budget	Proceed only if the primary outcome is measurable and the attainable sample can detect an economically worthwhile effect.

Period	Required deliverable	Investment decision
Weeks 7–12	Randomized exploration and a locked prospective policy comparison; models and parameters dated before evaluation	A validated simple-policy benefit can justify continuation. Without model benefit, omit the correlated model. Without measurable business benefit, stop or redesign within the remaining budget.
Months 4–6	Mature outcomes, agreed profit or cost reconciliation, invoice and collection, second deployment, recorded integration/support hours	Proceed to recurring product investment only if a customer pays, the benefit survives the full cost ledger, and a second account can be served economically.
After month 6	Repeat value after changing conditions, renewal evidence, and a bottom-up account list with acquisition economics	Expand distribution and product scope only as evidence supports them; marketplace and guarantee functions remain separate decisions.

Calendar dates do not override the outcome horizon. Returns or sufficient independent events may take longer than twelve weeks. A documented extension must have a remaining information objective, cost ceiling and stopping date. The relevant milestone is information acquired per dollar spent, rather than the number of model features completed.

7.4 Customer acquisition and defensibility

The first sale should be direct and specific: an evaluation of a known promotion or service-routing decision with a capped scope and a shared outcome definition. Likely introductions are through a relevant commerce agency, operations integrator, or existing institutional relationship. These are proposed channels; no partnership is claimed. The founder records the budget owner, present solution, data access, delivery effort, procurement steps and reason for rejection for every serious conversation.

After two implementations, the product should standardize the smallest recurring workflow: data connector, decision constraints, policy execution, measurement and reconciliation. If each new customer requires a different research project, pricing must support a consultancy or the target segment must narrow. A consultancy can produce profit, but it has different capacity and valuation economics from repeatable software revenue. Labeling custom labor as recurring software does not resolve that limitation.

Defensibility would come from permissioned outcome histories, task or product taxonomy, reliable integrations, accumulated experimental knowledge and a measurement process the customer's finance team trusts. The theorem itself is public. Data do not automatically compound across customers: permissions, confidentiality and distribution differences may require tenant-specific models. A lower-cost platform can integrate a similar optimizer, so the customer's recurring reason to buy must be measured and revisited.

8 What would justify a larger commitment

The research program has a clear hierarchy of evidence. Mathematical identification establishes what specified experiments can learn under a model. Held-out prediction assesses that model on new outcomes. A prospective policy experiment measures economic improvement. A paid and renewed contract measures whether a customer values the service after delivery costs. Trading requires the analogous progression through executable prices, fills, risk, capacity and realized capital use. None of these steps substitutes for the next.

The merchant proposal is the provisional first choice because its decision can be randomized and its value reconciled on a channel the customer controls. The opportunity remains conditional on traffic, data access, attainable effect size and acquisition economics. AI service routing can be preferable where enterprise access is stronger and absolute outcomes are verifiable. Prediction-market research has accessible technical interfaces, but no trading advantage follows from coherence alone. Its first commercial question is whether enough net executable edge remains after realistic costs.

Three research results would materially strengthen the case: prospective evidence that correlated substitution improves profit beyond standard offer experimentation; measured savings from complementary fallback providers after verification errors; and event-contract returns that survive settlement review, chronological holdouts and conservative execution. Until those observations exist, a broad exchange or a balance-sheet guarantee would add obligations faster than information. The next commitment should purchase the missing evidence in a narrow workflow with a paying counterparty.

A Minimum data contracts

The following schemas describe requirements for a future pilot. No customer or exchange dataset with these fields is supplied by this paper.

Merchant decisions. Record a unique decision identifier; pseudonymous assignment unit; UTC assignment and exposure times; eligible menu and offer variants; all displayed prices, stock and delivery terms; predecision features; policy version and assignment probabilities; selected action; order identifiers; and a complete outcome window, including no order. Join order-level collected revenue, discounts, cost of goods, shipping, fees, cancellations, refunds and return processing costs. Store both event time and availability time where they differ. Log exposures before observing conversion. Retain the allocation history when a customer returns. Keep model-development extracts separate from the final prospective measurement sample.

Event-contract decisions. Record venue and contract identifiers; exact rule text or retained permitted reference and its version; event-family membership; exhaustive outcome-state mapping; resolution time, source and exceptional-state treatment; market status; fee schedule and effective date; prices and quantities at every used depth level; exchange and local receive timestamps; feed sequence; trades; simulated or actual order lifecycle; cancellation delays; partial fills; paid fees; collateral; and final settlement cash flows. Simulation, actual order and settlement records must have distinct fields. A final report must not silently replace unsuccessful simulations with later better prices.

AI tasks. Record task identifier and organization; arrival time; permitted predecision features; provider, model and workflow version; supplied tools; routing probability; invoice cost; start and end time; timeout; output; verifier decision; blinded adjudication where available; later correction; human time; and downstream loss or a separately labelled unresolved outcome. A paired evaluation must record whether B saw A's output or diagnosis. This changes the meaning of the joint failure law.

B Questions for the first buyer meetings

The proposed interview asks for concrete operating evidence. Which promotion, inquiry or task-routing decision is made repeatedly? Who owns its budget and can approve a controlled change? What is the current rule and which product or internal team already supports it? Which outcomes and costs are recorded, including failures, no-purchases and returns? How much eligible volume occurs in a month? Can the parties reconcile a representative outcome sample to the accounting system? What decisions cannot be randomized, and why? What fee and implementation effort would be acceptable if the agreed benefit were measured?

A promising meeting ends with a named data owner, a feasible experiment, a costed implementation scope and a paid deliverable. The absence of any one does not prove that the entire opportunity is false, but it identifies the next uncertainty to resolve. A proposed conversion funnel of fifteen conversations to one pilot is a work target, not a forecasted sales conversion rate. No outreach has been undertaken for this paper.

C Commercial hypotheses and validation requirements

The commercial proposals concern model-based event trading, merchant offers, agent selection and guarantees. Their investment cases depend on the following distinctions and validation requirements.

1. Structural relationships among events are divided into exact payoff constraints and statistical dependence. Only the former give contractual lower bounds, and those bounds apply after acquiring the package.

2. The utility Jacobian is used as a sensitivity conditional on a chosen model. The mapping from order flow to latent information is a separate unproved and unvalidated component.

3. Discounts are actual interventions on offers. Their interpretation as known common-scale utility shifts requires additional assumptions, while a randomized policy comparison can assess profit without those assumptions.

4. Agent rankings are supplemented by absolute completion, cost, latency and severity data. A score-offset experiment does not identify those targets merely by recovering relative Gaussian performance.

5. Linear-in-catalog pair counts are reported together with repeated trials, conditioning requirements and the cost of new task regimes. There is no claim that a million-agent service can be rated cheaply from a few isolated comparisons per agent.

6. Company scale is expressed through collected account revenue, direct costs, acquisition effort and customer counts. Large market narratives and speculative valuation labels are replaced by conditions a pilot can test.

D Reproducibility and source boundaries

All numerical tables are generated by the accompanying economic calculation script from a single assumption file. The program checks probability normalization, the discount accounting, valid joint-failure probabilities, the sample-size formulas, exact implication payoffs and observation counts. Inputs and script hashes are stored with the resulting tables. These are deterministic calculations, not simulations of observed trading or customer behavior. Changing the assumptions changes the projections; no input is estimated from commercial data.

From the parent project directory, reproduce the calculations and document with

```
.venv/bin/python commercial/src/economics.py
latexmk -pdf -interaction=nonstopmode -halt-on-error \
  -cd commercial/main.tex
```

The accompanying source register identifies each external claim, its primary source and the date consulted. Live interface and fee descriptions may change. Historical results cited from other authors have not been independently replicated here. No claim of exhaustive novelty, commercial diligence, legal clearance or empirical validation is made. The original mathematics and application manuscripts are separate works and retain their own proof and review boundaries.

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